

EFFECTS OF NUTRIENT SOURCES ON FRUIT LYCOPENE AND SELECTED PHYTOCHEMICALS OF TOMATO (*Solanum lycopersicum* L.) IN OGBOMOSO, NIGERIA

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ABSTRACT

Tomato (*Solanum lycopersicum* L.) is a globally important vegetable crop valued for its nutritional and health-promoting properties, particularly its lycopene content. However, tomato fruit quality in Nigeria remains constrained by declining soil fertility and suboptimal nutrient management practices. This study evaluated the effects of different nutrient sources on fruit lycopene and selected phytochemicals of two tomato varieties (Roma VF and UC82B) in Ogbomoso, Nigeria. A field experiment was conducted at the Teaching and Research Farm, Ladoké Akintola University of Technology, Ogbomoso, Nigeria. The experiment was a 2 × 6 factorial combination in a Randomized Complete Block Design with three replications. The treatments were two tomato varieties (Roma VF and UC82B) and six fertilizer treatments: Control, 100% NPK (20-10-10), 100% *Tithonia* compost (10 t/ha), Supergro foliar fertilizer (1 L/ha), 50% NPK + 50% Compost, and 25% NPK + 75% Compost. Data were collected on proximate composition, mineral content, vitamin content, and lycopene content. Data were subjected to analysis of variance using SAS version 9.4 and means were separated using Duncan's Multiple Range Test at 5% probability level. For proximate composition, UC82B outperformed Roma VF across all parameters, recording crude protein (1.790%), crude fibre (0.694%), fat (0.245%), ash (0.354%), and moisture (87.858%). Foliar application consistently produced the highest crude protein (1.897%), crude fibre (0.742%), fat (0.283%), ash (0.415%), and moisture (87.858%). For vitamins, UC82B had vitamin C (40.728 mg/100g) and vitamin A (0.926 mg/100g), while foliar application produced the highest vitamin C (46.228 mg/100g) and vitamin A (1.020 mg/100g). Lycopene content was not significantly different between varieties. However, foliar fertilizer produced significantly higher lycopene (22.22 mg/100g) than the control (11.29 mg/100g). For farmers prioritizing nutritional quality and high lycopene content, UC82B variety with foliar application of Supergro (1 L/ha) is recommended in the study area.

Keywords: *Solanum lycopersicum*, *Tithonia diversifolia* compost, Supergro, lycopene, organo-mineral nutrition, antioxidant capacity.

INTRODUCTION

Tomato (*Solanum lycopersicum* L.) stands as one of the most widely cultivated and consumed vegetable crops globally, with significant contributions to human nutrition and dietary diversity.

The crop belongs to the Solanaceae family and originates from South America, though its cultivation has spread to virtually all tropical, subtropical, and temperate regions of the world (FAO, 2025). In Nigeria, tomato production plays a crucial role in household food security, income generation, and employment provision for millions of smallholder farmers. Despite its importance, the productivity of tomato in Nigeria remains far below its potential, primarily due to declining soil fertility, inadequate nutrient management practices, and the use of low-yielding varieties (Kaboré *et al.*, 2022). The nutritional significance of tomato cannot be overstated, as the fruit is an excellent source of vitamins, minerals, dietary fiber, and bioactive phytochemicals that confer numerous health benefits to consumers (Collins *et al.*, 2022).

Among the various bioactive compounds in tomato, lycopene has attracted considerable scientific attention due to its potent antioxidant properties and its association with reduced risk of chronic diseases, including certain cancers and cardiovascular disorders (Bin-Jumah *et al.*, 2022). Lycopene is a carotenoid pigment responsible for the characteristic red color of ripe tomato fruits, and its concentration in tomato is influenced by genetic factors, environmental conditions, and agronomic practices (Romdhane *et al.*, 2023). The growing consumer awareness of the health benefits associated with tomato consumption has increased the demand for nutritionally superior tomato varieties and production practices that enhance the content of beneficial phytochemicals (Lima *et al.*, 2022). Furthermore, recent comparative studies on *Capsicum* cultivars have demonstrated significant variability in nutritional, bioactive, and mineral profiles among varieties grown under similar conditions, emphasizing the importance of cultivar selection in determining fruit quality attributes (Adedayo *et al.*, 2026).

Fertilizer management constitutes one of the most critical agronomic interventions affecting tomato growth, yield, and fruit quality. Nitrogen, phosphorus, and potassium are the primary macronutrients required for optimal tomato production, and their availability significantly influences vegetative development, flowering, fruit set, and the accumulation of secondary metabolites (Jin *et al.*, 2022). Mineral fertilizers, particularly NPK, have been widely adopted in tomato production systems to address nutrient deficiencies and enhance productivity. However, the sole reliance on inorganic fertilizers has raised concerns regarding soil degradation, environmental pollution, and the long-term sustainability of production systems (Kadam and Mane, 2022). Organic amendments such as compost, manure, and biofertilizers have been recommended as alternatives or supplements to mineral fertilizers for improving soil health and enhancing fruit quality (Kabore *et al.*, 2022).

The application of *Tithonia* compost has gained attention in tropical agricultural systems due to its high nutrient content and its potential to improve soil fertility and crop productivity (Akanbi, 2002). *Tithonia diversifolia*, commonly known as wild sunflower, is an invasive weed species with exceptional nutrient scavenging capacity, and its biomass can be converted into high-quality compost for agricultural use. Studies have demonstrated that *Tithonia* compost can supply substantial amounts of nitrogen, phosphorus, and potassium to crops, while also improving soil physical and biological properties (Chabi *et al.*, 2024). Furthermore, the combined application of organic and inorganic fertilizers, through integrated nutrient management approaches, has been shown to synergistically improve crop performance and nutritional quality (Rocha *et al.*, 2023). Foliar fertilization represents another promising approach for enhancing tomato nutrition and fruit quality. Foliar applied nutrients are absorbed directly through the leaf surface, bypassing

soil constraints such as nutrient fixation, low availability, and root uptake limitations (Falsafi *et al.*, 2022). Supergro, a commercially available foliar fertilizer, has been reported to improve growth, yield, and phytochemical accumulation in various vegetable crops. The rapid absorption and efficient utilization of foliar applied nutrients can support the biosynthesis of secondary metabolites, including lycopene and vitamins, particularly during critical growth stages such as flowering and fruit development (Badea *et al.*, 2025).

The choice of tomato variety also plays a decisive role in determining the nutritional quality of fruits. Roma VF and UC82B are two determinate tomato varieties commonly cultivated in Nigeria, each possessing distinct morphological characteristics, disease resistance profiles, and suitability for different end-uses (Schmidt-Szantner *et al.*, 2022). Roma VF is characterized by oval-shaped fruits with thick flesh and low juice content, making it ideal for processing, while UC82B produces pear-shaped fruits with firm texture and good shelf life, suitable for both processing and fresh consumption (Sargsyan *et al.*, 2025). The varietal differences in lycopene content and phytochemical accumulation may interact with nutrient management practices, necessitating the evaluation of variety-specific fertilizer responses for optimizing nutritional quality.

Despite the considerable research on tomato fertilization, limited information exists on the combined effects of variety and diverse nutrient sources on lycopene and phytochemical accumulation in tomato fruits under Southwestern Nigeria conditions. The interaction between tomato varieties and nutrient sources may result in differential responses that have implications for designing variety-specific fertilizer recommendations aimed at enhancing nutritional quality. Moreover, the comparative performance of Tithonia compost and foliar fertilizer relative to conventional NPK application in improving fruit phytochemical content remains poorly understood. This study therefore evaluated the effects of different nutrient sources on fruit lycopene and selected phytochemicals of two tomato varieties in Ogbomoso, Nigeria

MATERIALS AND METHODS

Experimental Site

The experiment was conducted at the Teaching and Research Farm, Ladoké Akintola University of Technology (LAUTECH), Ogbomoso, Oyo State, Nigeria. The location lies at latitude 8°10'N and longitude 4°16'E above sea level.

Experimental Materials

Seeds of two tomato varieties, Roma VF and UC82B, were used for the study. The two varieties are commonly cultivated, consumed, and adapted to the western agroecological zones of Nigeria. The seeds were obtained from Irurun-Agbe Agro Allied Stores, Ogbomoso.

Tithonia compost was used and sourced from Right Choice Farm for the study. Prior to its use, samples were taken for the analysis of P, K, Ca, and Mg contents in the laboratory. Nitrogen (N) was determined using the Kjeldahl method, phosphorus (P) by Bray-1 method, potassium (K) by ammonium acetate method, while calcium (Ca) and magnesium (Mg) were analyzed using Atomic Absorption Spectroscopy (AAS). The nutrient contents were used to determine the equivalent quantity required to meet the fertilizer recommendation for tomato production.

Preparation of *Tithonia* Compost

Tithonia compost was prepared following the method described by Akanbi (2002). The compost was prepared from wild sunflower biomass and poultry manure. The poultry manure was sorted, and stones, feathers, metals, broken bottles, and other non-biodegradable materials were removed. The heap method of composting was adopted, while the wall of the compost bin was lined with black polythene sheets.

After collecting sufficient quantities of *Tithonia* biomass and poultry manure, the materials were laid in layers at a ratio of 3:1 (*Tithonia*: poultry manure) on a dry weight basis (Akanbi, 2002; Adediran *et al.*, 2003). Ventilation poles were inserted, and small holes were created for efficient aeration. Turning was carried out fortnightly, while pH, compost heap temperature, and ambient temperature were monitored weekly. The compost was monitored for eight weeks. At maturity, the compost was evacuated from the bin, air-dried, shredded, bagged in sacks, and stored under shade at room temperature until use. Samples were taken for chemical analysis of N, P, K, Ca, Mg, Fe, Zn, and Cu using standard methods according to IITA (2023).

Plant Material

Seeds of the two selected tomato varieties were obtained from Irorun-Agbe seed suppliers. The seeds were sown in sterilized nursery soil and raised for three weeks until the seedlings reached the 4-6 leaf stage. Uniform and healthy seedlings were transplanted into the field plots.

Mineral Fertilizer Used

The mineral fertilizer used was NPK 20-10-10, which is one of the readily available mineral fertilizers in the study area. The quantity applied depended on the results of the pre-cropping soil analysis and fertilizer recommendations for tomato production in Southwestern Nigeria.

Table 1: Characteristics of Tomato Varieties Used in the Study

Variety	Growth Habit	Fruit Characteristics	Major Uses	Special Attributes
Roma VF	Determinate	Oval-shaped fruits with thick flesh and low juice content	Processing and fresh consumption	Resistant to <i>Verticillium</i> and <i>Fusarium</i> wilt diseases
UC82B	Determinate	Pear-shaped fruits with firm texture and good shelf life	Processing and market	High yield potential and adaptability to tropical environments

Experimental Design and Treatments

The experiment comprised two tomato varieties: Roma VF (V1) and UC82B (V2). Six nutrient management treatments were evaluated, including a no-fertilizer control (F1), 100% NPK 20-10-10 applied at 300 kg/ha (F2), 100% compost at 10 tons/ha (F3), foliar fertilizer at 1 L/ha (F4), 50% NPK combined with 50% compost (F5), and 25% NPK combined with 75% compost (F6). The experiment was arranged as a 2 × 6 factorial combination, giving a total of 12 treatment combinations replicated three times. The experiment was laid out in a Randomized Complete Block Design (RCBD) with three replicates. Each plot measured 2 m × 2 m and contained 16 plants spaced at 0.3 m × 0.3 m. A buffer distance of 0.5–1 m was maintained between plots to

prevent nutrient crossover. Compost was applied at 10 t/ha and incorporated into the soil 2–3 weeks before transplanting.

NPK fertilizer was applied at 300 kg/ha to supply 60 kg N, 30 kg P₂O₅, and 30 kg K₂O per hectare as basal application at transplanting and as top-dressing during flowering and fruiting stages. Integrated treatments, F5 (50% NPK + 50% compost) and F6 (25% NPK + 75% compost), were applied according to the same schedule. Foliar fertilizer (F4) was applied as a 5% solution at weekly intervals from four weeks after transplanting until fruit set. Plants were staked and pruned to optimize growth, improve airflow, and minimize disease incidence.

Nutritional Quality Analysis

Lycopene Content

Lycopene extraction was carried out following the method described by Fish *et al.* (2002). Fresh tomato fruit samples were homogenized, and approximately 1 g of sample was extracted using a mixture of hexane, acetone, and ethanol in the ratio of 2:1:1. The mixture was shaken vigorously and allowed to separate into layers. The upper hexane layer containing lycopene pigment was collected, and absorbance was measured spectrophotometrically at 503 nm. Lycopene concentration was calculated using the appropriate extinction coefficient and expressed as mg/100 g fresh weight.

Vitamin C Content

Vitamin C content was determined using the 2,6-dichlorophenolindophenol titrimetric method as described by AOAC (2019). Fresh tomato juice extract was titrated against standardized dye solution until a light pink endpoint persisted for 15 seconds. Vitamin C concentration was calculated and expressed as mg ascorbic acid per 100 g fresh fruit sample.

Proximate Composition

Proximate composition (crude protein, crude fibre, fat, ash, and moisture) was determined using standard AOAC (2019) methods. Crude protein was determined by the Kjeldahl method, crude fibre by acid-base digestion, fat by Soxhlet extraction, ash by muffle furnace incineration at 550°C, and moisture by oven drying at 105°C until constant weight.

Mineral Content

Mineral elements (Na, K, Ca, P, Fe, Zn, and Mg) were determined using standard methods. Samples were digested using a tri-acid mixture (nitric acid, sulfuric acid, and perchloric acid). Sodium and potassium were determined using flame photometry, while calcium, magnesium, iron, and zinc were analyzed using Atomic Absorption Spectroscopy (AAS). Phosphorus was determined using the vanado-molybdate colorimetric method (IITA, 2023).

Statistical Analysis

All data collected were subjected to Analysis of Variance (ANOVA) appropriate for a factorial experiment arranged in a Randomized Complete Block Design (RCBD) using SAS version 9.4. Treatment means were separated using Duncan's Multiple Range Test (DMRT) at 5% level of probability ($p < 0.05$).

RESULTS

Effects of Variety and Fertilizer on Proximate Composition of Tomato Fruits

Table 2 presents the interaction effects of variety and fertilizer on proximate composition of tomato fruits. The interaction between variety and fertilizer was significant for all proximate parameters, indicating that the two varieties responded differently to fertilizer treatments for nutritional quality. For crude protein, UC82B under foliar application recorded the highest value (1.925%), which was statistically similar to UC82B under compost (1.850%) and Roma VF under foliar (1.870%). The lowest crude protein was recorded for Roma VF under the control (1.615%), which was statistically similar to UC82B under the control (1.640%).

For crude fibre, UC82B under foliar application recorded the highest value (0.765%), significantly higher than Roma VF under the control (0.610%) and UC82B under the control (0.605%). For fat content, UC82B under foliar application recorded the highest value (0.3050%), while Roma VF under the control recorded the lowest (0.1450%). For ash content, UC82B under foliar application recorded the highest value (0.4250%), while UC82B under the control recorded the lowest (0.2450%). For moisture content, UC82B under foliar application recorded the highest value (88.785%), while Roma VF under the control recorded the lowest (81.000%).

Table 2: Effects of Variety and Fertilizer on Proximate Composition of Tomato Fruits

Variety	Fertilizer	Crude Protein (%)	Crude Fibre (%)	Fat (%)	Ash (%)	Moisture (%)
Roma VF	Compost	1.770b	0.675c	0.225c	0.365c	86.800b
Roma VF	Control	1.615e	0.610e	0.145e	0.260e	81.000e
Roma VF	Foliar	1.870a	0.720b	0.260b	0.405b	86.930b
Roma VF	NPK	1.705d	0.640d	0.180d	0.325d	83.650d
UC82B	Compost	1.850a	0.740b	0.265b	0.400b	88.635a
UC82B	Control	1.640e	0.605e	0.170d	0.245e	81.675e
UC82B	Foliar	1.925a	0.765a	0.305a	0.425a	88.785a
UC82B	NPK	1.745c	0.665c	0.240c	0.345d	85.650c

Means with the same letters along the column are not significantly different at $p \leq 0.05$ using DMRT. 100% NPK: NPK 20-10-10; Foliar: Supergro (1 L/ha); 100% Compost: *Tithonia* Compost (10 t/ha).

Effects of Variety and Fertilizer on Vitamin Composition of Tomato Fruits

Table 3 presents the interaction effects of variety and fertilizer on vitamin composition of tomato fruits. The interaction between variety and fertilizer was significant for both vitamin C and vitamin A, indicating that the two varieties responded differently to fertilizer treatments for vitamin content. For vitamin C, UC82B under foliar application recorded the highest value (48.625 mg/100g), significantly higher than all other treatment combinations. Roma VF under the control recorded 31.690 mg/100g, while UC82B under the control recorded the lowest value (30.740 mg/100g). The intermediate values were recorded for UC82B under compost (44.820 mg/100g), Roma VF under foliar (43.830 mg/100g), Roma VF under compost (40.190 mg/100g), UC82B under NPK (38.725 mg/100g), and Roma VF under NPK (36.750 mg/100g).

For vitamin A, UC82B under foliar application recorded the highest value (1.0750 mg/100g), significantly higher than all other treatment combinations. The lowest vitamin A was recorded

for UC82B under the control (0.8050 mg/100g), which was statistically similar to Roma VF under the control (0.8150 mg/100g). The intermediate values were recorded for Roma VF under foliar (0.9650 mg/100g), UC82B under compost (0.9550 mg/100g), Roma VF under compost (0.9200 mg/100g), UC82B under NPK (0.8700 mg/100g), and Roma VF under NPK (0.8650 mg/100g).

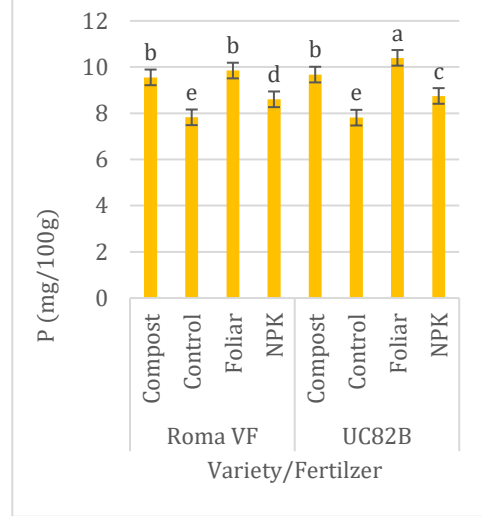
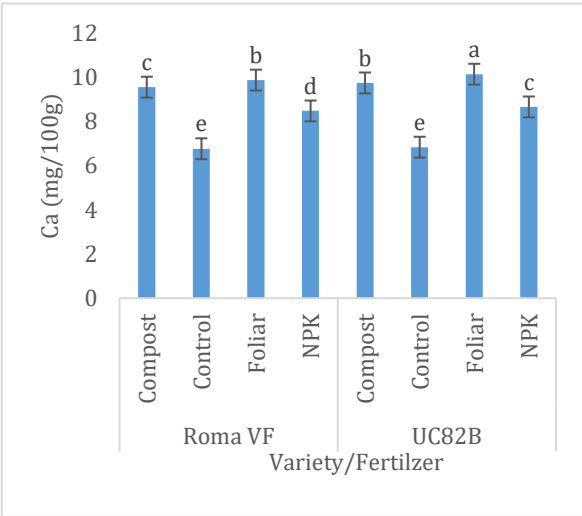
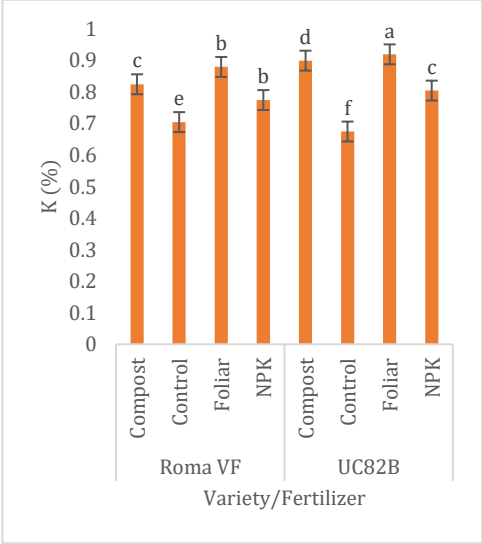
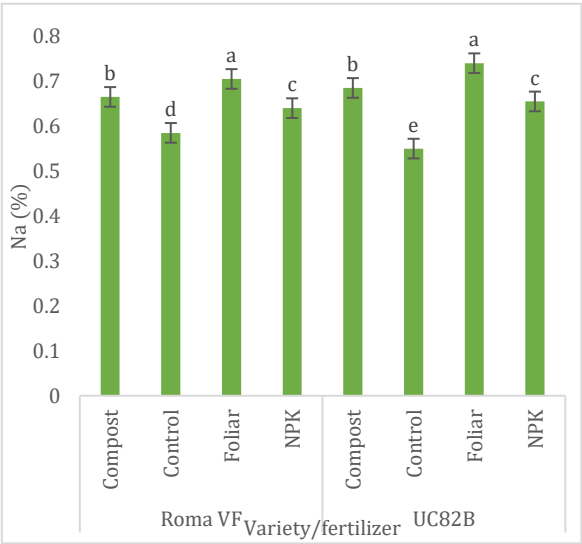
Table 3: Effects of Variety and Fertilizer on Vitamin Composition of Tomato Fruits

Variety	Fertilizer	Vitamin C (mg/100g)	Vitamin A (mg/100g)
Roma VF	Compost	40.190c	0.9200c
Roma VF	Control	31.690e	0.8150e
Roma VF	Foliar	43.830b	0.9650b
Roma VF	NPK	36.750d	0.8650d
UC82B	Compost	44.820b	0.9550b
UC82B	Control	30.740f	0.8050e
UC82B	Foliar	48.625a	1.0750a
UC82B	NPK	38.725c	0.8700d

Means with the same letters along the column are not significantly different at $p \leq 0.05$ using DMRT. 100% NPK: NPK 20-10-10; Foliar: Supergro (1 L/ha); 100% Compost: *Tithonia* Compost (10 t/ha).

Effects of Variety and Fertilizer on Mineral Composition of Tomato Fruits

Figure 1 presents the interaction effects of variety and fertilizer on mineral composition of tomato fruits. The interaction between variety and fertilizer was significant for all mineral parameters. For sodium, UC82B under foliar application recorded the highest value (0.7400%), while UC82B under the control recorded the lowest (0.5500%). For potassium, UC82B under foliar application recorded the highest value (0.9200%), while UC82B under the control recorded the lowest (0.6750%). For calcium, UC82B under foliar application recorded the highest value (10.1300 mg/100g), while Roma VF under the control recorded the lowest (6.7600 mg/100g). For phosphorus, UC82B under foliar application recorded the highest value (10.4000 mg/100g), while UC82B under the control recorded the lowest (7.8100 mg/100g). For iron, UC82B under foliar application recorded the highest value (1.5600 mg/100g), while UC82B under the control recorded the lowest (0.8400 mg/100g). For zinc, Roma VF under compost recorded the highest value (0.9350 mg/100g), while UC82B under the control recorded the lowest (0.6600 mg/100g). For magnesium, both Roma VF and UC82B under foliar application recorded the highest value (1.1250 mg/100g), while Roma VF under the control recorded the lowest (0.6700 mg/100g).



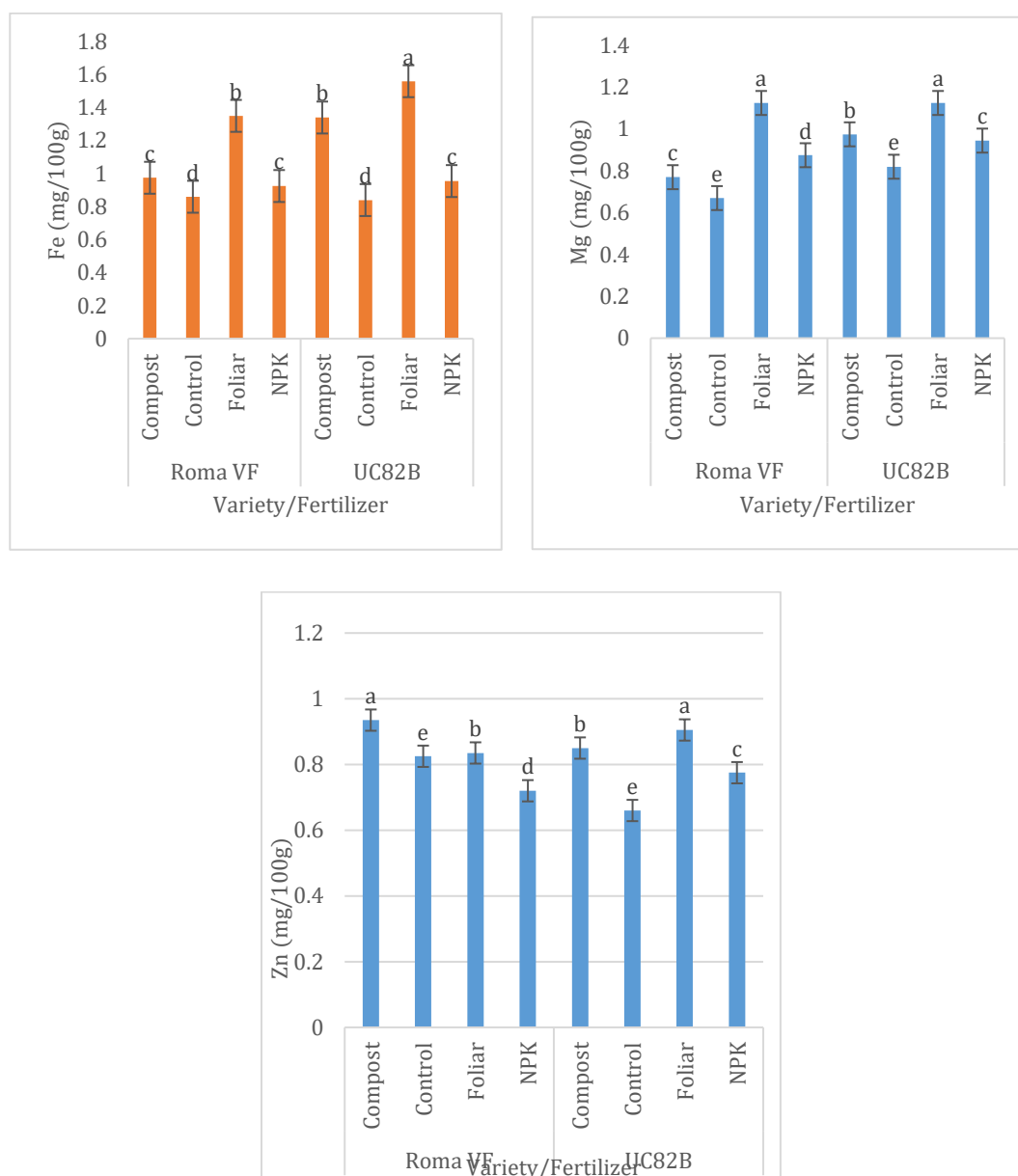


Figure 1: Effects of variety and fertilizer on mineral composition of tomato fruits

3.4 Effects of Variety and Fertilizer on Lycopene Content of Tomato Fruits

Table 4 presents the effects of variety and fertilizer on lycopene content of tomato fruits. For Roma VF, lycopene content ranged from 13.77 mg/100g at the control to 21.70 mg/100g at foliar application. The 100% compost treatment recorded 18.85 mg/100g, while 100% NPK recorded 16.76 mg/100g. For UC82B, lycopene content ranged from 8.80 mg/100g at the control to 22.74 mg/100g at foliar application. The 100% compost treatment recorded 20.77 mg/100g, while 100% NPK recorded 17.81 mg/100g. UC82B under foliar application recorded the highest lycopene content (22.74 mg/100g), which was statistically similar to Roma VF under foliar (21.70 mg/100g) and UC82B under compost (20.77 mg/100g). The lowest lycopene content was

recorded for UC82B under the control (8.80 mg/100g), which was significantly lower than all other treatment combinations.

Table 4: Effects of Variety and Fertilizer on Lycopene Content (mg/100g) of Tomato Fruits

Variety	Fertilizer	Lycopene (mg/100 g)
Roma VF	Control	13.77d
Roma VF	100% NPK	16.76c
Roma VF	100% Compost	18.85bc
Roma VF	Foliar	21.70ab
UC82B	Control	8.80e
UC82B	100% NPK	17.81c
UC82B	100% Compost	20.77ab
UC82B	Foliar	22.74a

Means with the same letters along the column are not significantly different at $p \leq 0.05$ using DMRT. 100% NPK: NPK 20-10-10; Foliar: Supergro (1 L/ha); 100% Compost: *Tithonia* Compost (10 t/ha).

DISCUSSION

Proximate Composition

The significant interaction between variety and fertilizer for all proximate parameters indicates that the nutritional quality of tomato fruits is influenced by the combined effects of genotype and nutrient management. UC82B under foliar application consistently recorded the highest values for crude protein (1.925%), crude fibre (0.765%), fat (0.305%), and ash (0.425%), demonstrating the superiority of this variety when combined with foliar nutrition for enhancing nutritional quality (Chabi *et al.*, 2024). The elevated crude protein content under foliar fertilization likely reflects the efficient absorption and utilization of nitrogen through foliar pathways, supporting amino acid and protein synthesis during fruit development (Badea *et al.*, 2025). Foliar application allows for rapid nutrient absorption directly through the leaf epidermis, bypassing soil-related constraints such as nutrient fixation and leaching, thereby ensuring efficient nutrient delivery during critical growth stages (Falsafi *et al.*, 2022). This mechanism is particularly important for nitrogen, which is a fundamental component of amino acids and proteins, and its availability directly influences protein synthesis in developing fruits (Jin *et al.*, 2022).

The significantly higher moisture content recorded for UC82B under foliar application (88.785%) compared to Roma VF under control (81.000%) suggests that foliar nutrition enhances water retention in fruits, which may influence texture and consumer acceptability (Lima *et al.*, 2022). The maintenance of adequate moisture content in tomato fruits is critical for maintaining turgor pressure, which affects fruit firmness and shelf life. Foliar application of nutrients, particularly potassium, has been shown to regulate stomatal conductance and water use efficiency, potentially leading to improved water status in developing fruits (Kulawik *et al.*, 2023). Furthermore, the enhanced moisture content under foliar treatment may reflect improved cellular hydration and membrane integrity, which are essential for maintaining fruit quality during storage and transportation (Kim and Imm, 2025).

The crude fibre content of tomato, while relatively low, contributes to dietary fiber intake, and the higher values observed under foliar application for UC82B (0.765%) compared to the control (0.605%) indicate that nutrient management can enhance this nutritional component (Rocha *et al.*, 2023). Dietary fibre plays important roles in human health, including improving digestive function, regulating blood sugar levels, and reducing cholesterol absorption. The enhancement of fibre content under foliar fertilization may be attributed to improved carbohydrate metabolism and cell wall synthesis, processes that are influenced by the availability of micronutrients such as boron and calcium (Wirkijowska *et al.*, 2023). These micronutrients, often present in foliar fertilizer formulations, are essential for cell wall formation and structural integrity, thereby contributing to increased fibre content in fruits (Teterycz and Sobota, 2023).

The fat content of tomato fruits increased with foliar application, with UC82B recording the highest value (0.305%), suggesting that foliar nutrition may influence lipid metabolism or storage in fruits (Wirkijowska *et al.*, 2023). Although tomato is not a major dietary source of fats, the presence of lipids in fruits is important for the absorption of fat-soluble vitamins and carotenoids, including lycopene. The enhancement of fat content under foliar fertilization may be related to improved availability of precursors for lipid biosynthesis, such as acetyl-CoA, which is derived from carbohydrate metabolism (Pathak and Sagar, 2023). Additionally, foliar-applied micronutrients such as magnesium and zinc are cofactors for enzymes involved in lipid metabolism, potentially explaining the observed increase in fat content (Chabi *et al.*, 2024).

The ash content, representing mineral accumulation, was also highest under foliar application, indicating that foliar nutrition enhances the uptake or retention of essential mineral elements (Teterycz and Sobota, 2023). Ash content is a reflection of the total mineral content of the fruit, and higher values indicate greater accumulation of essential minerals such as potassium, calcium, magnesium, and phosphorus. The enhanced mineral accumulation under foliar treatment may result from direct absorption of nutrients through leaf surfaces, as well as improved root uptake due to enhanced plant vigor and root development (Kadam and Mane, 2022). Furthermore, the presence of chelating agents in foliar fertilizer formulations may improve the bioavailability and translocation of micronutrients to developing fruits (Rocha *et al.*, 2023).

The superior proximate composition observed for UC82B across most treatments suggests that this variety has inherent genetic advantages for nutritional quality, although the magnitude of these advantages is influenced by fertilizer management. Genetic differences in nutrient uptake efficiency, translocation, and assimilation can lead to variations in fruit composition among varieties (Schmidt-Szantner *et al.*, 2022). The generally higher nutrient values under organic and foliar treatments compared to inorganic NPK alone suggest that integrated nutrient approaches may produce nutritionally superior fruits (Romdhane *et al.*, 2023). Organic amendments such as compost provide not only macronutrients but also micronutrients and organic matter that improve soil health and nutrient cycling, potentially leading to more balanced nutrient availability and enhanced fruit quality (Kadam and Mane, 2022). The moisture content range of 81.000% to 88.785% observed across treatments falls within the typical range for tomato fruits, though the higher values under organic and foliar treatments may indicate improved water retention and potential processing characteristics (Kim and Imm, 2025). These findings demonstrate that both variety selection and nutrient management can be strategically employed to enhance the nutritional quality of tomato fruits for different end-use applications. The

observed varietal differences in proximate composition are consistent with recent findings on Capsicum cultivars, which documented significant genetic variability in nutritional profiles among varieties grown under similar conditions (Adedayo *et al.*, 2026).

Vitamin Composition

The significant interaction between variety and fertilizer for both vitamin C and vitamin A indicates that the accumulation of these essential vitamins in tomato fruits is influenced by genotype and nutrient management. UC82B under foliar application recorded the highest vitamin C content (48.625 mg/100g), significantly exceeding all other treatment combinations, including Roma VF under foliar (43.830 mg/100g) and UC82B under compost (44.820 mg/100g). This finding aligns with reports that foliar fertilization enhances vitamin C biosynthesis, likely through improved nitrogen and potassium nutrition that supports ascorbic acid metabolism (Pathak and Sagar, 2023). Vitamin C (ascorbic acid) is synthesized in plants through the L-galactose pathway, which requires various enzymes and cofactors that are influenced by nutrient availability (Collins *et al.*, 2022). Potassium, in particular, plays a crucial role in ascorbic acid synthesis by activating enzymes involved in the pathway and maintaining cellular pH and redox balance (Kulawik *et al.*, 2023).

The vitamin C content of tomato fruits is particularly important for human health, as it serves as a potent antioxidant and contributes to immune function (Collins *et al.*, 2022). Vitamin C is involved in collagen synthesis, iron absorption, and protection against oxidative stress, making it an essential nutrient for human health. The substantial difference between foliar-treated and control treatments (48.625 vs 30.740 mg/100g for UC82B) demonstrates the magnitude of improvement achievable through appropriate nutrient management. This represents an approximately 58% increase in vitamin C content, which is nutritionally significant and could contribute substantially to daily vitamin C requirements (Lima *et al.*, 2022). The enhancement of vitamin C under foliar fertilization may also be related to improved micronutrient status, as micronutrients such as copper, zinc, and manganese are cofactors for enzymes involved in ascorbic acid metabolism (Chabi *et al.*, 2024).

For vitamin A, UC82B under foliar application also recorded the highest value (1.075 mg/100g), significantly higher than the control treatments (0.805 mg/100g for UC82B and 0.815 mg/100g for Roma VF). Vitamin A in tomato is primarily derived from provitamin A carotenoids, including beta-carotene, which are precursors to retinol in the human body (Bin-Jumah *et al.*, 2022). The enhancement of vitamin A under foliar fertilization suggests that nutrient management can influence carotenoid biosynthesis pathways, possibly through effects on ethylene production and ripening processes (Romdhane *et al.*, 2023). Carotenoid biosynthesis is regulated by various enzymes, including phytoene synthase and phytoene desaturase, which are influenced by nutrient availability, particularly nitrogen and potassium (Badea *et al.*, 2025). The role of potassium in activating these enzymes has been well documented, and foliar application of potassium-rich fertilizers such as Supergro may directly enhance carotenoid accumulation (Kulawik *et al.*, 2023).

The intermediate values recorded for compost treatments (40.190 mg/100g vitamin C and 0.920 mg/100g vitamin A for Roma VF) indicate that organic fertilization provides substantial vitamin enhancement compared to the control, though not as pronounced as foliar application (Kadam

and Mane, 2022). The slower release of nutrients from compost compared to foliar fertilizers may result in more gradual nutrient availability, which, while beneficial for overall plant nutrition, may not provide the concentrated nutrient supply needed for maximum vitamin accumulation during fruit development (Rocha *et al.*, 2023). Nevertheless, the significant enhancement of vitamins under compost treatment compared to the control demonstrates the value of organic amendments in improving fruit nutritional quality (Chabi *et al.*, 2024).

The superior vitamin content of UC82B compared to Roma VF under equivalent treatments suggests that this variety possesses genetic advantages for vitamin accumulation, possibly related to its determinate growth habit and fruit development characteristics (Schmidt-Szantner *et al.*, 2022). Genetic variation in vitamin biosynthesis pathways and nutrient use efficiency can lead to differences in vitamin accumulation among varieties (Sargsyan *et al.*, 2025). The vitamin C values recorded in this study (31.690 to 48.625 mg/100g) fall within the typical range for tomato fruits, though the upper values approach the levels found in some citrus fruits, highlighting the nutritional significance of well-managed tomato production (Wirkijowska *et al.*, 2023). The vitamin A content of tomato, while lower than green leafy vegetables, contributes meaningfully to dietary vitamin A intake, particularly in regions where provitamin A-rich foods are limited in the diet (Rocha *et al.*, 2023). The findings demonstrate that combining appropriate variety selection with optimal nutrient management, particularly foliar fertilization, can substantially enhance the vitamin content of tomato fruits, with direct implications for improving the nutritional quality of this widely consumed vegetable. Furthermore, the observed varietal differences in vitamin accumulation are consistent with recent findings on *Capsicum* cultivars, where significant genetic variability in bioactive compound profiles was documented among varieties grown under similar conditions (Adedayo *et al.*, 2026).

Mineral Composition

The significant interaction between variety and fertilizer for all mineral parameters reveals that the accumulation of essential minerals in tomato fruits is influenced by genotype, nutrient availability, and their interactions. UC82B under foliar application consistently recorded the highest values for sodium (0.740%), potassium (0.920%), calcium (10.130 mg/100g), phosphorus (10.400 mg/100g), iron (1.560 mg/100g), and magnesium (1.125 mg/100g), demonstrating the superior mineral accumulation capacity of this variety when combined with foliar nutrition (Kaboré *et al.*, 2022). The enhanced mineral content under foliar application can be attributed to the direct absorption of nutrients through leaf surfaces, which bypasses soil-related constraints such as nutrient fixation, leaching, and antagonistic interactions among ions (Falsafi *et al.*, 2022). Foliar fertilizers typically contain chelated forms of micronutrients that are readily absorbed and translocated to developing fruits, resulting in higher mineral concentrations (Badea *et al.*, 2025).

The enhanced potassium content under foliar application is particularly noteworthy, as potassium plays a crucial role in fruit quality, influencing color development, sugar accumulation, and shelf life (Kulawik *et al.*, 2023). Potassium is essential for enzyme activation, osmoregulation, and cation-anion balance in plant cells, and its adequate supply is critical for optimal fruit development and quality (Jin *et al.*, 2022). The high potassium content recorded for UC82B under foliar application (0.920%) compared to the control (0.675%) represents a 36% increase, which

could significantly improve fruit quality attributes such as sweetness, acidity balance, and post-harvest storage life (Pathak and Sagar, 2023). Furthermore, potassium is involved in the biosynthesis of carotenoids, including lycopene, by activating enzymes in the carotenoid pathway, which may explain the concurrent enhancement of both potassium and lycopene under foliar treatment (Romdhane *et al.*, 2023).

The calcium content under foliar application for UC82B (10.130 mg/100g) was substantially higher than the control (6.760 mg/100g for Roma VF), highlighting the importance of nutrient management in addressing calcium deficiency, which can lead to blossom-end rot and other physiological disorders (Li *et al.*, 2023). Calcium is essential for cell wall structure and membrane integrity, and its deficiency can result in reduced fruit quality and shelf life (Kadam and Mane, 2022). The enhancement of calcium content under foliar application suggests that direct application of calcium-containing foliar sprays can effectively supplement root uptake, particularly in soils with low calcium availability or high antagonistic ion concentrations (Chabi *et al.*, 2024). The higher calcium content in UC82B fruits compared to Roma VF under similar treatments may reflect genetic differences in calcium uptake efficiency or translocation patterns (Sargsyan *et al.*, 2025).

The phosphorus content of tomato fruits, while relatively low, contributes to energy metabolism and membrane integrity, and the enhancement observed under foliar application indicates improved phosphorus nutrition (Jin *et al.*, 2022). Phosphorus is a component of nucleic acids, phospholipids, and ATP, and its adequate supply is essential for various metabolic processes during fruit development (Wirkijowska *et al.*, 2023). The higher phosphorus content under foliar treatment (10.400 mg/100g for UC82B) compared to the control (7.810 mg/100g for UC82B) suggests that foliar application effectively supplements phosphorus nutrition, potentially improving fruit quality and nutritional value (Rocha *et al.*, 2023).

The iron content recorded under foliar application for UC82B (1.560 mg/100g) was nearly double that of UC82B under the control (0.840 mg/100g), demonstrating the potential of foliar nutrition to address iron deficiency, a common nutritional problem in tropical vegetable production (Chabi *et al.*, 2024). Iron is essential for chlorophyll synthesis, electron transport, and various enzymatic reactions, and its deficiency can result in chlorosis, reduced photosynthesis, and impaired fruit development (Kadam and Mane, 2022). The substantial enhancement of iron content under foliar application has significant implications for human nutrition, as iron deficiency anemia is a major public health concern, particularly in developing countries (Collins *et al.*, 2022). Tomato fruits with enhanced iron content could contribute meaningfully to dietary iron intake, especially in populations with limited access to animal-based iron sources (Bin-Jumah *et al.*, 2022).

The zinc content, with Roma VF under compost recording the highest value (0.935 mg/100g) and UC82B under the control recording the lowest (0.660 mg/100g), suggests that organic fertilization may be particularly effective for enhancing zinc accumulation in tomato fruits (Kadam and Mane, 2022). Zinc is a cofactor for numerous enzymes involved in protein synthesis, carbohydrate metabolism, and auxin biosynthesis, and its adequate supply is essential for optimal plant growth and fruit development (Teterycz and Sobota, 2023). The higher zinc content under compost treatment may be attributed to the presence of organic ligands that chelate zinc and enhance its bioavailability, as well as improved soil conditions that promote root uptake (Chabi

et al., 2024). The variability in zinc accumulation between varieties and treatments highlights the complexity of mineral nutrition and the importance of integrated nutrient management approaches (Rocha *et al.*, 2023).

The magnesium content was consistently highest under foliar application for both varieties (1.125 mg/100g), indicating the efficiency of foliar magnesium uptake and its role in chlorophyll synthesis and photosynthetic capacity (Pathak and Sagar, 2023). Magnesium is a central component of the chlorophyll molecule and is also a cofactor for enzymes involved in carbohydrate metabolism and protein synthesis (Kulawik *et al.*, 2023). The enhancement of magnesium content under foliar application may improve photosynthetic efficiency and carbohydrate accumulation in fruits, potentially contributing to improved fruit quality and yield (Badea *et al.*, 2025). The superior mineral composition observed for UC82B under foliar application aligns with the findings of Romdhane *et al.* (2023), who reported that integrated nutrient management enhances mineral accumulation in tomato fruits. The generally higher mineral values under organic and foliar treatments compared to inorganic NPK alone suggest that these approaches facilitate more balanced and efficient mineral acquisition by tomato plants (Rocha *et al.*, 2023). The notable enhancement of iron and calcium under foliar application has implications for the nutritional value of tomato as a dietary source of these essential micronutrients, particularly in populations at risk of deficiency (Bin-Jumah *et al.*, 2022). These findings are consistent with recent research on Capsicum cultivars, which demonstrated significant variability in mineral profiles among varieties and highlighted the potential for targeted nutrient management to enhance mineral content in vegetable crops (Adedayo *et al.*, 2026).

Lycopene Content

The substantial treatment effects revealed by the effects demonstrate that nutrient management exerts a strong influence on lycopene content in tomato fruits. Foliar fertilizer produced the highest lycopene content (22.22 mg/100g averaged across varieties), significantly exceeding the control (11.29 mg/100g), representing an approximately 97% increase. This remarkable enhancement of lycopene under foliar fertilization aligns with findings that foliar applied nutrients, particularly potassium and micronutrients, can stimulate carotenoid biosynthesis during fruit ripening (Romdhane *et al.*, 2023). The physiological basis for this enhancement likely involves the role of potassium in activating enzymes involved in the carotenoid biosynthesis pathway, including phytoene synthase and phytoene desaturase, which are critical for lycopene accumulation (Badea *et al.*, 2025). Potassium is also involved in the regulation of ethylene biosynthesis, which plays a key role in fruit ripening and carotenoid accumulation (Kulawik *et al.*, 2023).

The compost treatments also produced substantial lycopene enhancement, with UC82B under compost recording 20.77 mg/100g and Roma VF recording 18.85 mg/100g, compared to the control. This finding supports the use of organic amendments for improving fruit quality, as the slow release of nutrients and improved soil conditions may support more balanced and sustained nutrient supply during fruit development (Chabi *et al.*, 2024). Organic amendments such as Tithonia compost not only supply macronutrients but also improve soil physical and biological properties, enhance nutrient cycling, and provide micronutrients and organic compounds that

may influence carotenoid biosynthesis (Akanbi, 2002). The presence of humic substances in compost may also enhance nutrient uptake and translocation to fruits, potentially contributing to increased lycopene accumulation (Kadam and Mane, 2022).

The lowest lycopene content was recorded for UC82B under the control (8.80 mg/100g), demonstrating the critical importance of nutrient supply for lycopene biosynthesis. Nutrient deficiency, particularly of potassium and nitrogen, can impair carotenoid biosynthesis by reducing the availability of precursors and cofactors, as well as by affecting overall plant vigor and fruit development (Wirkijowska *et al.*, 2023). The lycopene values observed in this study (8.80 to 22.74 mg/100g) are within the range reported for various tomato cultivars, though the higher values achieved under foliar fertilization approach those of some high-lycopene varieties (Bin-Jumah *et al.*, 2022). The lack of significant varietal differences in lycopene content suggests that the genetic potential for lycopene accumulation is similar between Roma VF and UC82B, though the numerical superiority of UC82B under most treatments (22.74 vs 21.70 mg/100g at foliar application) may indicate a slight advantage (Kulawik *et al.*, 2023).

The finding that lycopene content responded more strongly to fertilizer management than to varietal differences has practical implications, suggesting that nutrient management can be strategically employed to enhance this valuable phytochemical regardless of the variety grown (Jin *et al.*, 2022). The strong positive response of lycopene to foliar fertilization, combined with the enhancement of other quality parameters including vitamins and minerals, makes foliar application a particularly attractive strategy for producing nutritionally superior tomato fruits for fresh consumption and processing (Falsafi *et al.*, 2022). The potential health benefits associated with higher lycopene consumption, including reduced risk of cardiovascular disease and certain cancers, underscore the importance of production practices that enhance lycopene content in tomato fruits (Collins *et al.*, 2022). Furthermore, the observed variability in lycopene content among treatments and varieties is consistent with recent findings on bioactive compound profiles in *Capsicum* cultivars, which demonstrated significant genetic and environmental influences on carotenoid accumulation (Adedayo *et al.*, 2026).

CONCLUSION

This study demonstrated that nutrient sources and tomato varieties significantly influenced the nutritional quality parameters of tomato fruits. UC82B with foliar fertilizer application produced nutritionally superior fruits with enhanced proximate composition, mineral content, vitamins, and lycopene. Lycopene content increased substantially under foliar fertilization compared to the control. For farmers prioritizing nutritional quality and high lycopene content, UC82B variety with foliar application of Supergro is recommended. Integrated nutrient approaches combining organic and inorganic sources offer intermediate benefits for quality. The findings underscore the importance of variety-specific nutrient management strategies for optimizing the nutritional quality of tomato fruits in Southwestern Nigeria.

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